ENGINEERING & DESIGN DIVISION

MECHANICAL DESIGN • 3D CAD MODELLING • 2D DRAFTING • PRODUCT DEVELOPMENT • REVERSE ENGINEERING • SHEET METAL DESIGN • DFM

From concept to manufacturing-ready engineering.

1. ABOUT CRECER GRANDE

Crecer Grande is an engineering and industrial services venture focused on supporting manufacturing and engineering businesses with practical, technically sound and commercially viable solutions. We combine engineering capability, industrial experience and a flexible technical network to support customers across design, manufacturing, industrial components, maintenance, quality systems and business support.

Our objective is to work as an extension of our customers' engineering teams by providing dependable technical support based on actual project requirements.

2. ABOUT THE DIVISION

Crecer Grande - Engineering & Design Division provides mechanical engineering, product development, CAD modelling and technical documentation services for industrial and manufacturing applications.

We support projects starting from a basic requirement, concept, sketch, existing component or legacy drawing and convert them into accurate, manufacturing-ready engineering information.

PROJECT TYPES

- New product development
- Existing product modification
- Custom machine components
- Sheet-metal products
- Fabricated assemblies
- Jigs & fixtures

ENGINEERING OUTPUTS

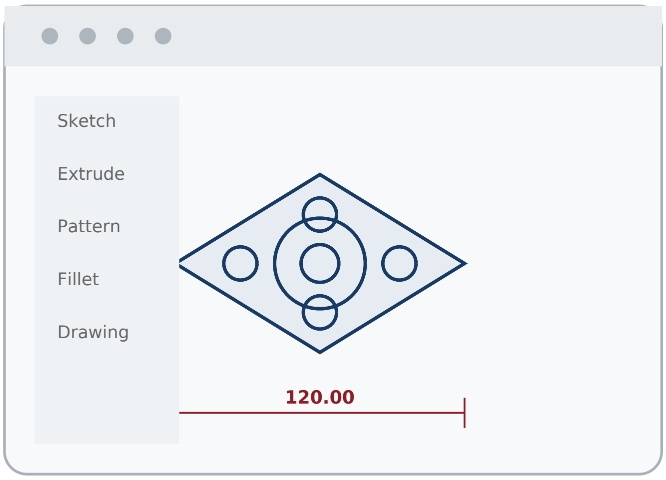
- 3D models & assemblies
- Manufacturing drawings
- Reverse-engineered data
- BOMs & technical documents
- Design modifications
- Production-ready packages

3. OUR CAPABILITIES

Mechanical Design & Product Development

From Requirement to Engineering Solution

- Concept development
- Mechanical component design
- Product design
- Machine component design
- Industrial equipment design
- Design modification
- Existing-product improvement
- Custom engineering solutions
- Prototype design support
- Design development from customer specifications



We work with customers to understand the function, application, operating environment, manufacturability and commercial requirements before developing the solution.

3D CAD Modelling

Accurate Digital Models for Engineering & Manufacturing

MODELLING CAPABILITIES

- Individual components
- Mechanical assemblies
- Sheet-metal components
- Fabricated structures
- Machine parts
- Equipment layouts
- Exploded assemblies
- Design modifications

INPUTS WE CAN USE

- Existing drawings
- Hand sketches
- Customer concepts
- Photographs + dimensions
- Sample components
- Existing CAD data
- Design specifications

4. 2D DRAFTING & MANUFACTURING DRAWINGS

Crecer Grande prepares detailed engineering drawings suitable for manufacturing, inspection and documentation. Our objective is to provide engineering information that is clear, practical and usable on the shop floor.

DRAWING TYPES

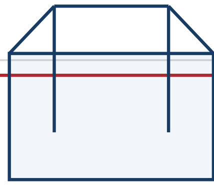
- Part drawings
- Assembly drawings
- Fabrication drawings
- Machining drawings
- Sheet-metal drawings
- GA drawings
- Detail drawings
- Installation drawings

DRAWING DETAILS

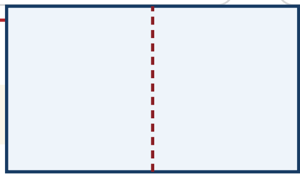
- Dimensions
- Tolerances & fits
- Material specifications
- Surface finish
- Welding information
- Manufacturing notes
- Part identification
- Revision information

DOCUMENT SUPPORT

- BOM references
- Drawing lists
- Revision updates
- Legacy conversion
- Drawing corrections
- Standardisation
- As-built drawings
- Modification drawings



5. SHEET METAL DESIGN



Engineering for Fabrication

- Enclosures
- Machine guards
- Covers
- Cabinets
- Brackets
- Mounting plates
- Panels
- Frames
- Trays
- Industrial sheet-metal assemblies

Design support can include 3D sheet-metal modelling, bend development, flat-pattern development, fabrication drawings, bend details, assembly detailing and manufacturing-feasibility review. Designs can be developed considering downstream processes such as laser cutting, bending, welding, machining and assembly.

6. ASSEMBLY DESIGN

Our assembly-design services support mechanical assemblies, fabricated assemblies, machine subassemblies, equipment structures, component integration, mounting arrangements and interface development.

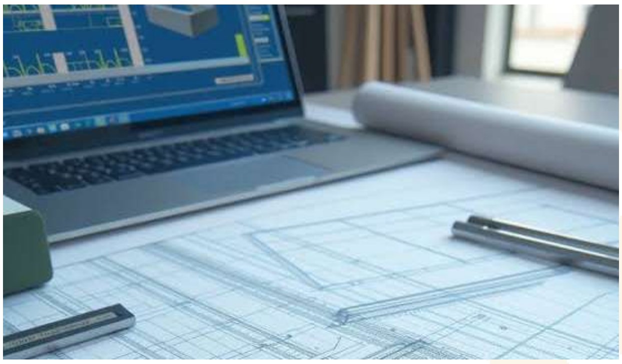
ASSEMBLY DELIVERABLES

- Exploded views
- Assembly sequences
- Part identification
- Bills of Materials
- Interface dimensions
- Installation references

WHY IT MATTERS

- Clearer assembly intent
- Reduced interface errors
- Structured documentation
- Simpler production handover
- Better change control

7. REVERSE ENGINEERING



From Existing Component to Engineering Data

When original drawings or CAD files are unavailable, Crecer Grande can recreate engineering information from an available component or assembly.

Measurement & Study

Functional Assessment

3D CAD Model

2D Manufacturing Drawing

Material / Specification Data

Typical applications include obsolete machine components, parts without drawings, imported machine parts, damaged components, legacy equipment and existing fabricated assemblies.

8. DESIGN FOR MANUFACTURING

A technically correct design is not always a commercially efficient design. Crecer Grande can support customers in reviewing designs from a manufacturing perspective.

DFM REVIEW

- Simplify components
- Reduce unnecessary machining
- Material optimisation
- Standardise features
- Reduce number of parts
- Fabrication-friendly changes
- Machining-friendly changes
- Assembly simplification

BALANCE THE DESIGN

- Function
- Manufacturability
- Quality
- Cost
- Maintainability
- Repeatability

9. DESIGN MODIFICATION & VALUE ENGINEERING

Existing products often require improvement rather than complete redesign. We can assist with cost reduction, weight reduction, material-substitution studies, component simplification, design upgrades, failure-related design improvement, replacement of obsolete components, standardisation and manufacturing-process improvement.

VALUE ENGINEERING FOCUS

Improve performance and manufacturability while reducing avoidable cost, complexity and production risk.

10. JIGS, FIXTURES & TOOLING DESIGN

- Welding fixtures
- Machining fixtures
- Assembly fixtures
- Inspection fixtures
- Drilling jigs
- Locating arrangements
- Clamping arrangements
- Holding fixtures
- Custom shop-floor tools
- Production aids

EXPECTED BENEFITS

- Improved repeatability
- Higher productivity
- Reliable positioning
- Operator convenience
- Inspection consistency

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11. BOM & ENGINEERING DOCUMENTATION

Crecer Grande can support engineering departments with structured technical documentation. This is particularly useful where designs have developed over time but engineering documentation has not been maintained systematically.

BOM

- Bill of Materials
- Part lists
- Assembly BOMs
- Component identification

DRAWING CONTROL

- Drawing lists
- Revision updates
- Standardisation
- Legacy conversion

SPECIFICATIONS

- Technical specs
- Material notes
- Drawing references
- Engineering records

12. DRAWING CONVERSION & LEGACY DATA SUPPORT

We can convert hand drawings, old paper drawings, scanned drawings, PDF drawings, customer sketches and legacy CAD data into updated 2D CAD drawings, 3D models, manufacturing drawings and revised technical documentation.

13. DESIGN REVIEW & ENGINEERING SUPPORT

For customers who already have designs, Crecer Grande can provide an independent engineering review before manufacturing.

REVIEW CHECKS

- Drawing completeness
- Dimensional consistency
- Assembly interfaces
- Manufacturing feasibility

DOCUMENT CHECKS

- Basic tolerance review
- Material specification review
- Drawing/BOM consistency
- Design-change verification

14. HOW WE WORK

Customer Requirement → **Requirement Study** → **3D CAD / Design** → **Design Review** → **2D Detail + BOM** → **Customer Approval**

Where required, the project can transition directly to Crecer Grande Advanced Manufacturing & Components for production support.

15. WHAT YOU CAN SEND US

Existing Drawing

For modification, correction or conversion.

Sketch

A hand sketch with available dimensions can be the starting point.

Existing Component

For measurement, reverse engineering and redevelopment.

Photograph

For initial assessment when accompanied by available dimensional/application information.

Existing CAD Model

For modification, development or drawing creation.

Product Concept / Requirement

Explain the function and objective; our team can help define an engineering approach.

16. INDUSTRIES WE SUPPORT

- General engineering
- Industrial machinery
- Sheet-metal manufacturing
- Fabrication
- CNC / VMC machining
- Automation
- Material handling
- Conveyor systems
- Paper & process industries
- Steel & metal industries
- Renewable energy
- Machine builders / OEMs
- Manufacturing MSMEs

17. WHY CRECER GRANDE

Engineering-Led

Focus on function and application, not only drawing production.

Design + Manufacturing Understanding

Engineering decisions consider downstream manufacturing realities.

Flexible Engagement

Suitable for one drawing, a full assembly or ongoing engineering support.

Reverse Engineering

Existing components can be converted into reusable engineering data.

Manufacturing Connection

Designs can transition directly into production support when required.

Practical Industrial Focus

Solutions are built around real manufacturing and maintenance needs.

18. OUR VALUE PROPOSITION

CONCEPT. DESIGN. DETAIL. DELIVER.

Whether you need:

- A new component designed
- A 3D CAD model
- Manufacturing drawings
- A product modified
- A sheet-metal assembly developed
- An existing part reverse engineered
- A jig or fixture designed
- Old drawings digitised
- A BOM prepared
- Ongoing engineering support

Crecer Grande can provide the engineering support required to convert your requirement into clear, manufacturing-ready technical information.

CONTACT US

CRECER GRANDE

Engineering & Design Division

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HAVE AN ENGINEERING REQUIREMENT?

Send us your drawing, sketch, CAD model, photograph, sample component or requirement for assessment.